Authorization systems can make this situation worse by requesting attributes needed for dynamic authorization behind the scenes without providing people the ability to give meaningful consent. Flipping that around, when people don’t know how policy decisions were made, they may have difficulty giving consent to attribute sharing and default to oversharing to avoid having to repeatedly make a request. Again, SSI was designed to help with this, and we’ll see in chapter 15 how SSI can aid in solving the consent problem in authorization systems.

2.2.6 The scale problem

Because digital identities underpin almost every digital interaction, organizations rely on them to be well-architected and carefully managed. At the same time, getting digital identity right is difficult. The scale problem arises from needing to manage identity well across a growing number of services, organizations, and digital interactions. Most organizations under-resource their identity infrastructure, which often leads to the scale problem at some point.

The emergence of dynamic authorization and organizations’ increasing reliance on it for critical operations only adds to the scalability issue. As we've discussed, dynamic authorization relies on information about the principals and resources referenced in policies. As organizations adopt a zero-trust model, scrutinizing every access request before granting authorization, the burden on identity systems intensifies. Without a scalable identity system, deploying dynamic authorization successfully in your organization is challenging.

2.2.7 How identity problems affect authorization

The challenges we’ve covered—proximity, interoperability, flexibility, privacy, consent, and scale—don’t just make identity management difficult. They also directly impact authorization, making it harder to reliably apply policies, retrieve accurate attributes, or even correctly identify who (or what) is making a request.